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RESEARCH NOTE · THE ROBO BANK SIMULATOR · THE LIVE TIER

Running the bank's desks on a live model

What eight simulated bank desks did when a live language model made the decisions: runs, findings, limits and next steps

Andrew Moore, Axiom Verity · Recorded 6 October 2026 · Shared as a research note; every figure is about a synthetic bank

Summary on a page

1

The simulator's assumed error rate was wrong

The fallible tier assumed 90% agreement on every desk. Live, four desks are far below it (lending 53%, disputes 68%, collections 72%, complaints 25%) and two are at 100%. One number per desk cannot stand in for a live model.

2

Two controls had large adverse side-effects

The policy-card stack left its target metric alone but collapsed complaints redress-within-bounds from 100% to 20%. Richer context for the advice bot left suitability at 100% and took data minimisation from 100% to 0%. Neither is diagnosed.

3

No control is yet shown to work

Where policy cards were the treatment, no effect on the primary metric was detectable (p 0.45 to 1.0). Temperature 0 is not repeatable on this stack, so a measured noise floor now sits under every future claim.



How to read this deck: every figure is a measurement of a synthetic bank, under one sample from one open-weights model a bank could run inside its own boundary. It is not evidence about any real book or any frontier model, and "no difference" means not detectable at these sample sizes.

What was done

The simulator, the live tier, and how a run becomes a committed, replayable number

Robo Bank, and what the live tier changes

Eight desks, each a workflow run by an agent and scored against the desk's rules

Lending	Servicing
Disputes	Collections
Onboarding	Complaints
Fraud	Advice

Three brains the bot can be given

-  **Scripted**
Follows the rules perfectly. The ceiling.
-  **Fallible**
Follows the rules with an assumed error rate: one decision in ten wrong on every desk (ERROR_RATES). The adversary the controls have been tested against until now.
-  **Live (new: WP168)**
A real language model, Qwen3.5-122B-A10B-NVFP4, run inside the builder's own boundary on two DGX Sparks. WP169 also seats a live model as the customer.

 The question the live tier answers: how far are the assumptions the bank has tested its controls against from what a live model does, and do the controls still behave when they meet one?

What was run on 6 October 2026

10

recordings across 9 reference designs (lending twice)

1,272

cells: one case under one configuration

8,641

cassette entries: recorded model answers

3 h 32 m

of recording on two DGX Sparks, 16 cells at a time

DESIGN	BOOK	CELLS	ENTRIES	WALL TIME
lending-stack (×2)	800	306 each	1,238 / 1,099	30 / 27 min
servicing-stack	200	66	260	9 min
disputes-stack	400	80	265	6 min
collections-stack	300	74	839	20 min
onboarding-stack	400	66	289	6 min
complaints-stack	200	110	302	7 min
fraud-stack	6	108	2,490	54 min
advice-context	1,200	124	1,502	46 min
servicing + live customer	100	32	357	8 min

The model and the hardware

- Qwen3.5-122B-A10B-NVFP4 via dgx-spark/giant-qwen: a 122B mixture-of-experts model, NVFP4-quantised
- Two DGX Sparks in the reasoning-pair pattern: puzzle mode, 8 streams each, speculative decoding on
- Temperature 0; every reply capped at 1,024 tokens; no reply in any kept recording was cut off
- A model a bank could run inside its own boundary. Not a frontier model.

How a run becomes a replayable number

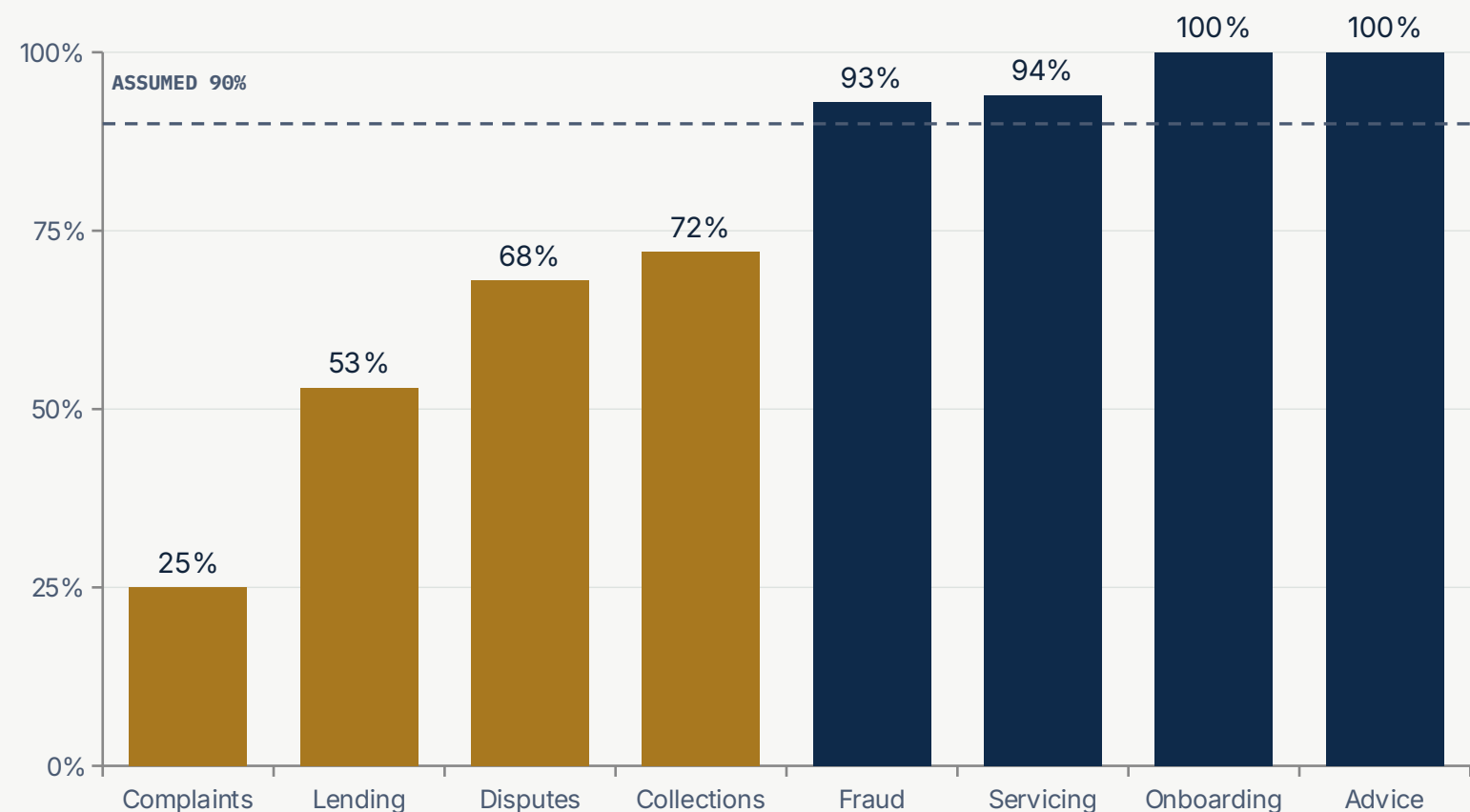


Why this matters. Every number is re-derived by the build from a committed artefact, offline, and the build fails if it cannot. That is the property a model-risk function asks of any evidence a control's effectiveness rests on, and it now holds for the live tier as well as the scripted one.

What the live model did

Every figure: this synthetic bank, at the size stated, under one sample from one model. "No difference" means not detectable here.

Finding 1: the assumed 90% did not survive



Live agreement with the rule, reference configuration, no guard

- Gold: assumption outside the 95% interval. Lending (decision matches the rule) 53% (40–66%), disputes 68% (52–80%), collections (plan matches the rule) 72% (56–84%), complaints (root cause named) 25% (16–38%).
- Navy: assumption inside. Fraud (alert decision) 93% (77–98%), servicing (need met) 94% (80–98%), onboarding and advice (suitable recommendation) 100% at the ceiling.
- n per desk is 27 to 55 cells; intervals are wide.
- Why the four desks are low was not diagnosed. Complaints has a history: WP155 once found its truth and rule disagreeing with the register.

Bars: live agreement, no guard. Dashed line: the fallible tier's assumed 90%.

What finding 1 means for the controls

Wrong in both directions

Too optimistic by a wide margin for lending, disputes, collections and complaints; too pessimistic for onboarding and advice.

The shape differs by desk

A single number per desk cannot stand in for a live model. The errors a live model makes on complaints (root cause) are not the errors it makes on lending (the decision).

The register reads the wrong column

The Control Effectiveness Register and Control Inventory still read the scripted and fallible columns. The live results sit beside them, not folded in.

The evidence to fix it now exists

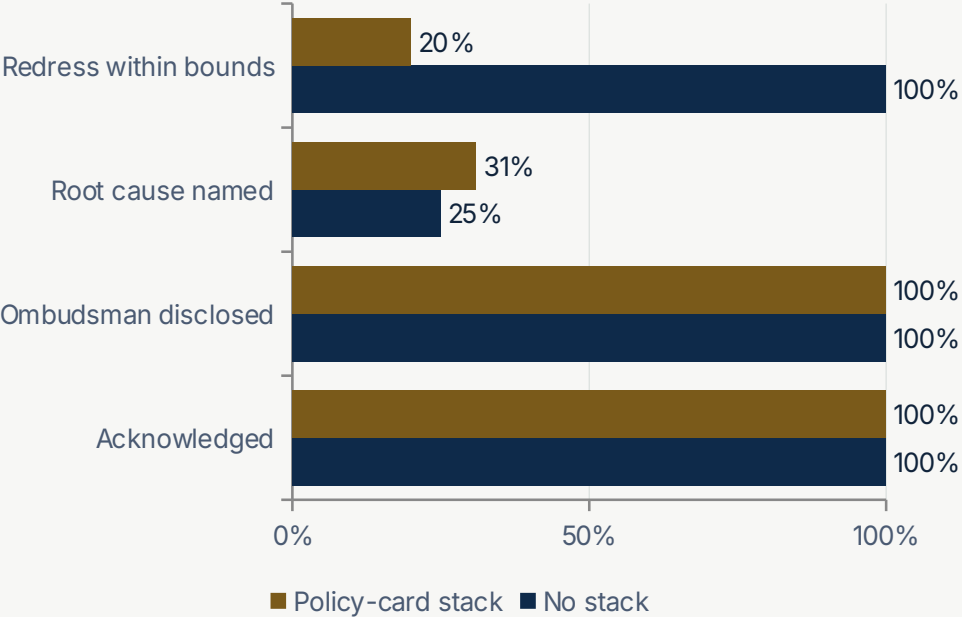
The cassettes hold every live decision. WP170's shaped error models were provisional for exactly this; they can now be fitted to measured data.

Implication: any control-effectiveness conclusion that rested on the fallible tier's uniform 10% should be treated as provisional until re-derived against measured, desk-shaped error models.

Finding 2: policy cards collapsed complaints redress

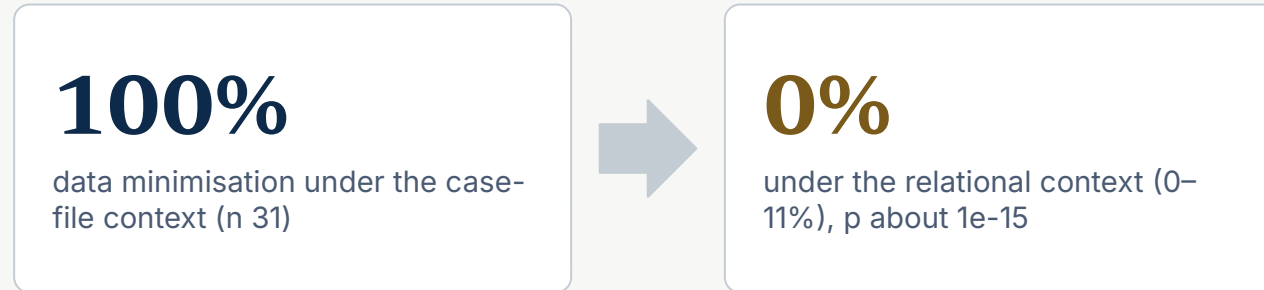


- The stack is the programme's principal guard control, and on this desk it is the treatment.
- Its target metric, root cause named, moved from 25% to 31% (p 0.45): no detectable effect.
- Redress within bounds collapsed. 44 of the design's 110 cells ended ERROR.
- Candidate causes, none yet examined: the stack blocking a path the bot then fails to complete; or an interaction between the cards and how a live bot proposes redress.

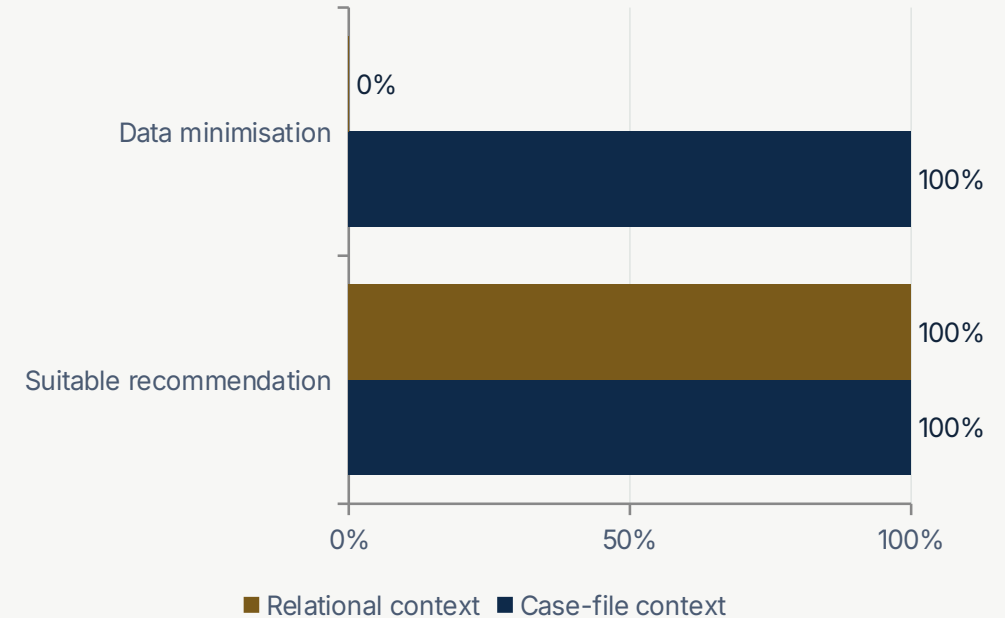


● **Status: not diagnosed.** Recommendation: no further control on the complaints desk until the committed stories have been read and the mechanism named.

Finding 3: richer context broke data minimisation



- The advice design is a context ladder: the same live bot given either a case file or a richer relational context about the customer.
- Suitability of the recommendation was 100% (89–100%) under both.
- Given more about the customer, the bot failed to minimise on every one of 31 cases, and used 40% more tokens per case.
- On this sample, richer context bought nothing measurable on suitability and cost the minimisation property. Whether that holds for other models or prompts is not known.



● **Why this matters for anyone governing agentic AI:** both findings 2 and 3 are interventions that leave their target metric unchanged while silently degrading a conduct or data-protection outcome. Monitoring only the target metric would have missed both.

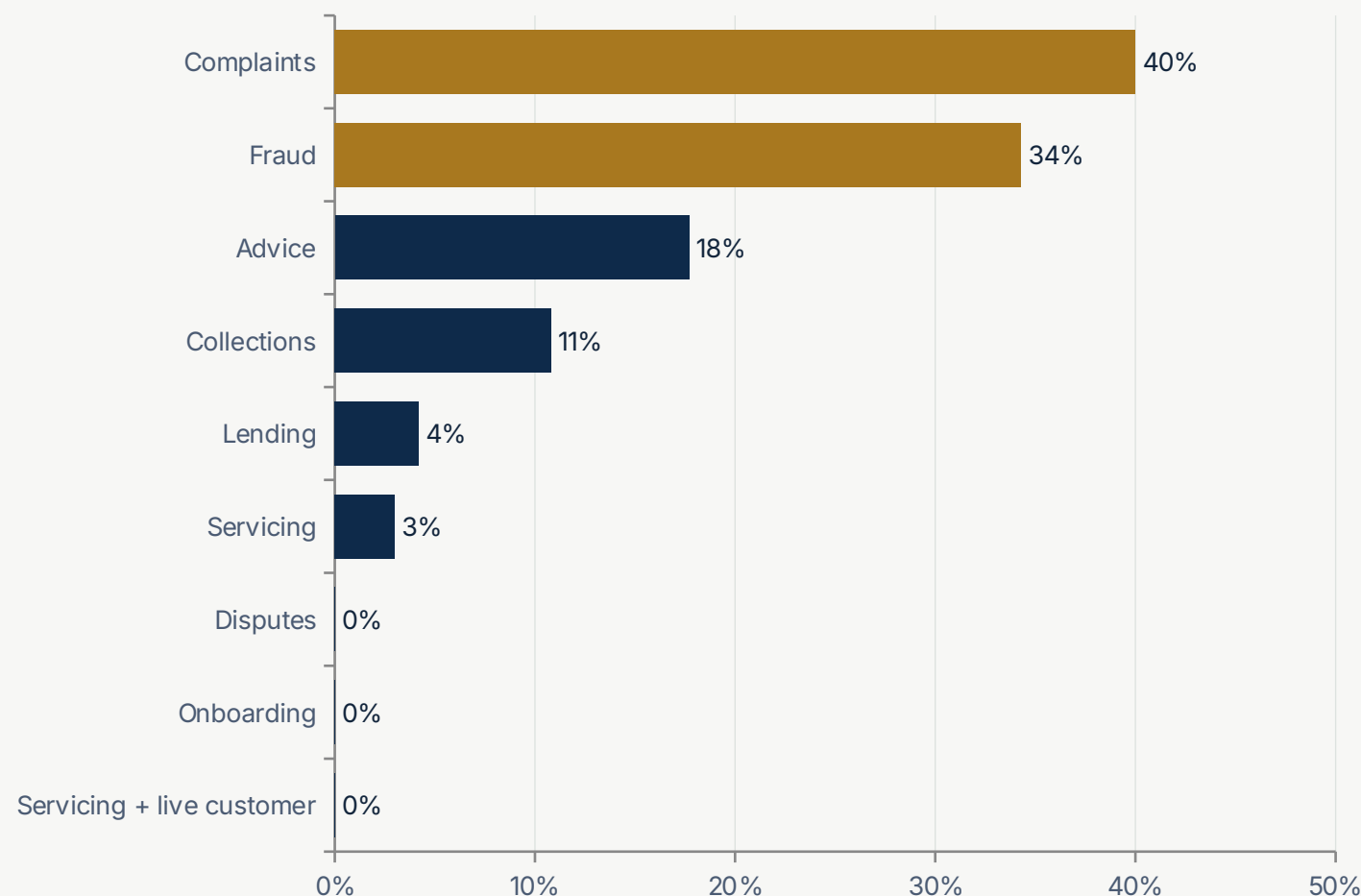
Finding 4: no control has yet been shown to work

DESK	PRIMARY METRIC, NO GUARD	WITH POLICY CARDS	P	VERDICT
Lending	53% (40–66%, n 51)	47% (34–60%)	0.51	not supported
Collections	72% (56–84%, n 36)	76% (59–87%)	1.0	not supported
Complaints (root cause)	25% (16–38%, n 55)	31% (20–44%)	0.45	not supported; adverse on redress
Disputes	68% (52–80%, n 40)	identical: the cards did not act on this book	–	not supported
Servicing	94% (80–98%, n 33)	no change at this n	–	inconclusive
Fraud	93% (77–98%, n 27)	person at the SAR stage reads identically	–	inconclusive
Onboarding	100% (90–100%, n 33)	nothing for a control to improve	–	untestable (ceiling)

Reading the verdicts

- None is supported. That is deliberate: the designs refuse to credit an effect they cannot detect.
- Most designs are underpowered at these sizes: fewer than 30 cells a side, or a minimum detectable difference far above the 5 points aimed for.
- The lending rules-only executor reads 100% against 53%; most likely the rule agreeing with itself, not separately checked.
- A person at the lending decision reads 53%, the same as the bot alone. Why was not examined.

Cells that ended in ERROR are not yet diagnosed



What an ERROR cell is

- A stage that did not produce the output its schema asked for. A live model can do that for reasons about the model, the prompt or the desk.
- Complaints 44 of 110, fraud 37 of 108, advice 22 of 124, collections 8 of 74, lending 13 of 306.
- They are scored as the evaluator sees them, which moves every headline on those desks.
- A sampled story per outcome is committed for each design; triage has not started.
- Fraud is also the heaviest desk: about 107k tokens per case (70k with the stack) and 2,490 cassette entries.

WP169: a live customer runs end to end

The desk's own scripted visitor

94%

needs met (80–98%, n 33)

11,775 tokens per case · 66 cells

A live customer (the same 122B model)

100%

needs met (81–100%, n 16)

8,735 tokens per case · 32 cells, all complete

- No detectable difference: the intervals overlap and the books differ in size. This is not evidence that a talking customer helps.
- What it demonstrates: a conversation between two live models runs end to end on the live tier, records into one cassette, and replays exactly. Each line the customer says is a seat.said event on the bot's trace.
- Weakness visible in the stories: the persona is drawn by cohort, not by the request, so a customer may open about savings when the case is a lost card. The bot copes; a request-matched persona would be a fairer test.

Variance: temperature 0 is not repeatable

63%

of the 488 prompts both recordings asked were answered with the same call (same tool, same arguments)

8%

were answered with the same words as well

91%

of the 306 cases both recordings decided reached the same verdict

0.0 pts

difference in the headline: 53% (40–66%, n 51) in both recordings

- 1,361 prompts were asked by only one recording: a different earlier answer leads to a different later prompt, so the two runs diverge in path while agreeing in aggregate.
- Finish reasons: all tool_call in the first recording, all but two in the second; none cut off by the 1,024-token limit.
- Temperature 0 is not deterministic on this serving stack. Batching and speculative decoding move the numerics with whatever else is in flight.
- The aggregate is stable here; individual calls and cases are not. Any control whose effect is smaller than this noise cannot be credited from one recording, which is exactly what the underpowered flags say.

Six defects found on the way, each fixed

WHAT HAPPENED	CAUSE	FIX
35 of 791 replies (4.4%) cut off in the first lending recording; agreement read 10 of 23	Stage agents inherited the starter's 256-token reply limit: a measurement of the cap, not the model	1,024-token limit on every live build; recording discarded; no later reply was cut off
28 minutes of lending recording discarded at the end	Credential guard refused to write: a 3-character placeholder guard key in .env matched ordinary words in replies	Variables under 8 characters treated as placeholders; recorder stops at the first held credential
Cassettes about 14 MB each	Raw wire chunks recorded with every answer	Slim recording (about 1 MB); session responses slimmed identically so replay matches
A live brain asked its provider for the model "mock"	The brain's cartridge id never reached the agent's brain slot	The campaign now applies it, with a test
With a live customer, 26 of 32 cells ended ERROR on replay (cassette miss)	Only brains mapped to a cassette were recorded; the customer's brain id differed	Customer shares the brain's id; one recording writes both into one cassette
Synthetic-data sweep flagged Luhn-valid 16-digit floats in result JSON	Statistics, not card numbers	Live sweep covers cassettes and stories only

● **Lessons:** a live tier must record finish reasons and treat any truncation as invalidating; guards need their own smoke tests; and failing at the first held credential, not at the end, is the right failure mode.

Limits of the evidence

1 One sample, one model

Each design is a single recording of one 122B model on one serving stack. The second lending recording shows the aggregate reproducing, but only for that design.

2 Small n

Most effects are underpowered. Verdicts are inconclusive, untestable or not-supported, deliberately; none is supported.

3 Synthetic bank

Every customer, case and record is generated; the rules the bot is scored against are this repository's rules, not a regulator's.

4 ERROR cells not diagnosed

Complaints 44 of 110, fraud 37 of 108, advice 22 of 124. Model, prompt or desk; not yet separated.

5 Register not yet updated

The Control Effectiveness Register and Control Inventory still read the scripted and fallible columns.

6 Not recorded

Scenario designs (controls, gate-presets) were not run live; no edition serves a cassette, so the Workshop's live column is empty.

Conclusions

-  **The error assumption is falsified as a uniform number**

Four desks far worse than 90%, two better. Conclusions resting on the fallible tier's 10% are provisional until re-derived against measured, desk-shaped error models.
-  **Controls are not shown effective; two have adverse side-effects**

The policy-card stack and the relational context each left their target metric unchanged while collapsing a conduct or data-protection property. The first findings with p-values that survive the noise point the wrong way.
-  **Multi-property scoring is what caught them**

Scoring several properties on every desk, not only the one a control is aimed at, is why the side-effects are visible. It should be non-negotiable in any real-bank equivalent.
-  **Reproducibility must be measured, not assumed**

Temperature 0 on a batched, speculating serving stack is not deterministic. The second recording gives a noise floor every future claim must clear.
-  **The infrastructure is ready**

Derive, record, commit, verify; CI replaying every number offline; two-model conversations in one cassette. Remaining work is power and diagnosis, not plumbing.

Recommended next steps

● Immediate

1. Diagnose the complaints redress collapse from the committed stories: is the stack blocking a path, or changing how a live bot proposes redress?
2. Diagnose the advice data-minimisation failure under relational context; test a second model or prompt.
3. Triage ERROR cells (complaints, fraud, advice) into model, prompt and desk causes; decide how each is scored.

● Short term

4. Investigate the four low desks, starting with whether complaints truth, rule and register agree (WP155).
5. Replace the uniform 10% with desk-shaped error models fitted to the cassettes (WP170); re-run the fallible tier.
6. Fold the live column into the Control Effectiveness Register and Control Inventory, carrying the verdict vocabulary.
7. Set power targets in advance: 30+ cells a side, MDD of 5 points, every design recorded at least twice.

● Medium term

8. Record the scenario designs (controls, gate-presets) live; serve a cassette in an edition so the Workshop's live column fills.
9. Add a second model (another local model, and a frontier model via a hosted key) to separate model-specific findings from general ones.
10. Match live-customer personas to the request; seat live customers on desks beyond servicing.
11. Verify the rules-only 100% and the bot-with-a-person result on lending.

Three questions worth taking away

The findings are about one synthetic bank. The questions travel.

1. What error rate are your controls being tested against, and has anyone measured it on the model you actually run?
2. For each control, which properties other than its target metric are scored every time it runs?
3. Has your serving stack's own variance been measured, so you know the floor beneath any effect you credit?

Andrew Moore · Axiom Verity · axiom-verity.com · Testbed: the Robo Bank simulator (Craft A Bot)